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Adjuvant mitotane in low-risk adrenocortical carcinoma: A reconstructed IPD analysis of the ADIUVO randomized and observational studies

Fabiano Flauto [a,*], Giuseppe Neola [a,b], Filippo Vitale [a], Vincenzo Damiano [a]

[a] Department of Clinical Medicine and Surgery, University of Naples Federico II, Via Sergio Pansini, 5, Naples 80131, Italy
[b] Portsmouth Hospitals University NHS Trust, Portsmouth, United Kingdom

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ABSTRACT

Background: The benefit of adjuvant mitotane after complete resection of low-risk adrenocortical carcinoma (ACC) remains uncertain. The ADIUVO randomized trial was underpowered due to low event rates. We conducted a reconstructed individual patient data (IPD) analysis to provide a comprehensive reassessment of treatment effects.

Methods: IPD were reconstructed from published Kaplan-Meier curves of the ADIUVO randomized trial and its parallel prospective observational cohort using the Guyot algorithm. Overall survival (OS) and recurrence-free survival (RFS) were analysed using Cox proportional hazards models, restricted mean survival time (RMST), and accelerated failure time (AFT) models. Pooled analyses were stratified by study design. Exposure-response analyses evaluated therapeutic mitotane levels ($\geq$14 mg/L) and early target attainment.

Results: The reconstructed randomized cohort included 91 patients (18 RFS events; 19 deaths). In intention-to-treat analysis, no clear evidence of a benefit with mitotane was observed for RFS (HR=0.99, 95%CI 0.39–2.50; p = 0.99) or OS (HR=0.98, 95%CI 0.40–2.42; p = 0.97). RMST differences at 60 months were negligible (-0.05 months for RFS; −0.19 months for OS). In the observational cohort (n = 95), no significant differences were observed for RFS (HR=0.90, 95%CI 0.32–2.47) or OS (HR=0.75, 95%CI 0.31–1.84). Pooled analysis (n = 186; 39 OS events; 33 RFS events) likewise showed no clear evidence of improved OS (HR=0.86, 95%CI 0.45–1.61) or RFS (HR=1.05, 95%CI 0.53–2.08). Exposure-based analyses did not demonstrate a consistent survival advantage associated with therapeutic drug levels or early target attainment, although these analyses were limited by small sample sizes.

Interpretation: Across complementary modelling approaches, no clear evidence of a clinically meaningful survival benefit with adjuvant mitotane was observed. While small effects cannot be definitively excluded, the overall consistency of findings suggests that any potential benefit is likely marginal. Active surveillance therefore represents an evidence-aligned management strategy in this population.

1. Introduction

Adrenocortical carcinoma (ACC) is a rare and aggressive endocrine malignancy, with an incidence of fewer than two cases per million individuals per year (Fassnacht et al., 2020; Fassnacht et al., 2018]. Its clinical course is heterogeneous. While many patients present with advanced or rapidly progressive disease, a substantial subset, particularly those with early-stage, low-grade tumours, experience indolent progression and a low risk of recurrence after complete surgical resection (Bilimoria et al., 2008). Identifying these low-risk patients and tailoring postoperative management accordingly has become a central objective in contemporary ACC care.

Mitotane, an adrenolytic derivative of DDT, has long been the cornerstone of systemic therapy for ACC (Else et al., 2014). Its activity in advanced disease is supported by randomised and large prospective studies (Fassnacht et al., 2012; Megerle et al., 2018). However, its role as adjuvant therapy after complete resection remains controversial (Terzolo et al., 2007; Grubbs et al., 2010). Mitotane has a narrow therapeutic window, requires prolonged time to achieve effective plasma concentrations, and is associated with substantial chronic toxicity, including neurocognitive impairment, gastrointestinal intolerance, adrenal insufficiency, and impaired quality of life (Haak et al.,

* Corresponding author.
E-mail address: fabiano.flauto@unina.it (F. Flauto).

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1994). In patients with a favourable baseline prognosis, such toxicity can only be justified by clear evidence of meaningful clinical benefit.

In high-risk ACC, defined by advanced ENSAT stage, high Ki67 index, vascular invasion, or incomplete resection, retrospective series and pooled analyses suggest a potential benefit from adjuvant mitotane (Berruti et al., 2010). In contrast, for patients with low-risk disease (ENSAT stage I-II, R0 resection, Ki67 ≤10%), evidence remains limited. In this subgroup, treatment decisions have historically relied on extrapolation from high-risk cohorts and expert consensus rather than direct randomised evidence (Araujo-Castro et al., 2025).

The ADIUVO trial was specifically designed to address this gap. This open-label, phase 3 randomised clinical trial (RCT) compared adjuvant mitotane with active surveillance in patients with completely resected, low-risk ACC. 91 patients were randomised, and recurrence-free survival (RFS) was the primary endpoint, with overall survival (OS) as a secondary endpoint (Terzolo et al., 2023).

At five years, RFS was 79% (95%CI 67–94) in the mitotane group and 75% (95%CI 63–90) in the surveillance group, corresponding to a hazard ratio (HR) of 0.74 (95%CI 0.30–1.85). Five-year OS was 95% (95%CI 89–100) with mitotane and 86% (95%CI 74–100) with surveillance, with an HR of 0.46 (95%CI 0.08–1.92) (Terzolo et al., 2023). Although point estimates numerically favoured mitotane, confidence intervals were wide and compatible with substantial benefit, no effect, or even potential harm. The trial therefore did not provide conclusive evidence supporting routine adjuvant therapy in this population (Terzolo et al., 2023).

The study was prematurely discontinued due to slow accrual and an unexpectedly low event rate. After nearly a decade of recruitment, only 91 patients were enrolled and 18 recurrences observed, far below the 66 events originally planned to ensure adequate statistical power. Therefore, treatment effect estimates were imprecise, and the trial was underpowered to confirm or exclude clinically relevant differences (Terzolo et al., 2023).

To complement the randomised component, the ADIUVO investigators established a parallel prospective observational cohort enrolling 95 patients under similar eligibility criteria. Treatment allocation in this cohort was determined by physician choice, reflecting real-world practice (Terzolo et al., 2023). While valuable, such data are inherently susceptible to confounding and selection bias and cannot be interpreted causally without careful modelling.

Both the randomised and observational components provided published Kaplan-Meier (KM) curves and risk-set tables for OS and RFS, as well as subgroup curves according to mitotane plasma concentrations and time to therapeutic levels. However, the individual patient-level data (IPD) have not been made publicly available, and the randomised and observational datasets have never been formally synthesised.

Moreover, although exploratory exposure analyses were reported in the original publication, they were limited by small subgroup sizes and conventional modelling approaches. A unified, patient-level reassessment of exposure-response relationships, incorporating time-to-target analyses and modern survival frameworks, has not been performed.

Advances in survival methodology now allow reconstruction of IPD from published KM curves. The algorithm described by Guyot et al. enables accurate estimation of individual event and censoring times using digitised curves, numbers at risk, and total event counts (Guyot et al., 2012). This approach has been validated and widely applied in oncology, permitting rigorous reanalysis when original IPD are unavailable (Tierney et al., 2007; Wei et al., 2015; Liu et al., 2021).

Building on these developments, we conducted a comprehensive reconstruction and reanalysis of the entire ADIUVO programme. By integrating data from the randomised trial, the prospective observational cohort, and the exposure substudies within a unified analytical framework, we aimed to provide the most precise and clinically relevant assessment to date of adjuvant mitotane in low-risk ACC. In addition to re-estimating treatment effects for RFS and OS, we specifically sought to clarify whether achieving or rapidly attaining therapeutic mitotane concentrations confers measurable benefit in this favourable-risk population.

2. Methods

2.1. Study design and data sources

This study is a secondary analysis of reconstructed IPD derived from the ADIUVO RCT and its parallel prospective observational cohort. As the original IPD were not publicly available, time-to-event data were reconstructed from published KM survival curves using established digitisation and reconstruction methods.

KM curves were digitised using WebPlotDigitizer (version 5.0). The time and survival axes were calibrated using predefined reference points (0 and maximum follow-up for time; 0% and 100% for survival probability). Between approximately 150 and 300 coordinate pairs per curve were extracted to ensure adequate resolution of early events, plateau phases, and long-term survival patterns. Numbers at risk at reported time points were manually transcribed from the original publication and incorporated into the reconstruction process.

The following survival datasets were analysed: OS and RFS in the randomized intention-to-treat (ITT) population; OS and RFS in the prospective observational cohort; OS and RFS according to mitotane plasma concentration (target ≥14 mg/L versus non-target <14 mg/L); and RFS according to time to attainment of therapeutic mitotane concentrations (≤3 months versus >3 months), applying a landmark approach where appropriate.

All randomized analyses were conducted according to the ITT principle, including all patients as originally allocated. Per-protocol population analyses from ADIUVO trial were not included in the present study to avoid post-randomization selection bias.

2.2. IPD reconstruction

IPD were reconstructed using the algorithm described by Guyot et al., which derives individual event and censoring times from digitised KM coordinates combined with reported sample sizes, total event counts, and numbers at risk (Guyot et al., 2012). Reconstructed survival curves were visually and numerically compared with the published KM curves to assess fidelity. Agreement was high, with minimal discrepancies in estimated survival probabilities and event counts across time points. Total numbers of reconstructed events closely approximated those reported in the original publication, with minor discrepancies attributable to the reconstruction process, particularly for endpoints with very low event counts. Published KM curves are provided in Supplementary Table 1.

Notably, the reconstructed HR for OS in the randomized cohort differed from the published estimate. This discrepancy likely reflects the extremely low number of OS events and the inherent instability of KM-based IPD reconstruction under sparse-event conditions, where small variations in event timing can substantially influence hazard estimates.

2.3. Statistical analysis

For each dataset, Cox proportional hazards models were fitted with treatment (mitotane vs observation) as the primary covariate. HRs with 95%CIs and Wald p-values were reported. Log-rank tests were used to compare survival distributions. For pooled analyses, randomized and observational datasets were combined, and treatment effects were estimated using stratified Cox models with study (RCT vs observational cohort) as a stratification factor. Wald p-values were derived from Cox proportional hazards models, while log-rank tests provided non-parametric confirmation of survival differences. Given the absence of covariates and the low event rate, both approaches yielded virtually identical results. All analyses were conducted in R (version 4.4.2).

Given the non-randomized nature of the observational cohort, pooled analyses were not intended to provide causal estimates of treatment effect. Treatment allocation in the observational setting was determined by physician choice and may reflect unmeasured clinical factors, introducing potential selection bias. Therefore, pooled results should be interpreted as a complementary assessment of the consistency and robustness of findings across different study designs rather than as causal inference.

2.4. Restricted mean survival time

Restricted mean survival time (RMST) was used as an absolute measure of treatment effect, defined as the area under the KM curve up to a prespecified time horizon (τ). RMST was estimated at 60 and 90 months, selected to cover the majority of observed follow-up without requiring extrapolation beyond available data. Between-group differences in RMST were estimated using the survRM2 package in R. RMST was computed only when the chosen truncation time (τ) was less than or equal to the minimum of the maximum observed follow-up times across comparison groups; otherwise, the estimate was considered non-estimable at that time horizon.

2.5. Accelerated failure time models

Accelerated failure time (AFT) models were fitted as a complementary parametric approach to quantify treatment effects on survival time. The time ratio (TR) represents the multiplicative change in expected survival time associated with treatment, with values greater than 1 indicating longer survival under mitotane and values less than 1 indicating shorter survival. Weibull, log-normal, and log-logistic distributions were evaluated for each dataset. In pooled analyses, study (RCT vs observational cohort) was included as a fixed covariate. Model fit was assessed using the Akaike Information Criterion (AIC), with lower values indicating better fit**.**

2.6. Censoring and follow-up

For each dataset and treatment arm, the number and proportion of censored observations were calculated. Median follow-up time was estimated using the reverse KM method, which treats censoring times as events to provide an unbiased estimate of follow-up duration. Censoring patterns were compared descriptively between treatment groups to evaluate potential imbalances that could influence hazard-based or RMST-based effect estimates.

3. Results

3.1. Randomised trial: RFS and OS in ITT

In the reconstructed ITT population of the randomized ADIUVO trial (n = 91; 18 RFS events), no evidence of a treatment effect was observed. Event counts were identical in both groups (9 recurrences in each arm). Cox regression yielded a HR of 0.99 (95%CI 0.39–2.50; Wald p = 0.987), indicating virtually identical recurrence hazards between mitotane and surveillance (Fig. 1). The log-rank test confirmed the absence of any difference (p = 0.986).

Restricted mean survival time analyses corroborated these findings. At 60 months, the absolute RMST difference between mitotane and surveillance was −0.045 months (95%CI; −6.85–6.76; p = 0.99). At 90 months, the difference was 0.065 months (95%CI; −12.49–12.62; p = 0.99). These values correspond to differences of only a few days and are clinically negligible.

Overall, the reconstructed ITT analysis suggests no detectable relative or absolute benefit of adjuvant mitotane on recurrence-free survival.

In the reconstructed ITT dataset of the randomised ADIUVO trial (n = 91), 19 deaths were observed, with 9 events in the mitotane arm (n = 45) and 10 in the observation arm (n = 46).

Cox regression analysis showed no evidence of a survival benefit associated with adjuvant mitotane. The HR for OS was 0.98 (95%CI; 0.40–2.42; Wald p = 0.97), indicating virtually identical hazard rates between the two groups. Consistently, the log-rank test showed no difference in survival distributions (p = 0.96) (Fig. 2).

RMST analyses confirmed the absence of clinically meaningful differences. At 60 months, the absolute difference in RMST between

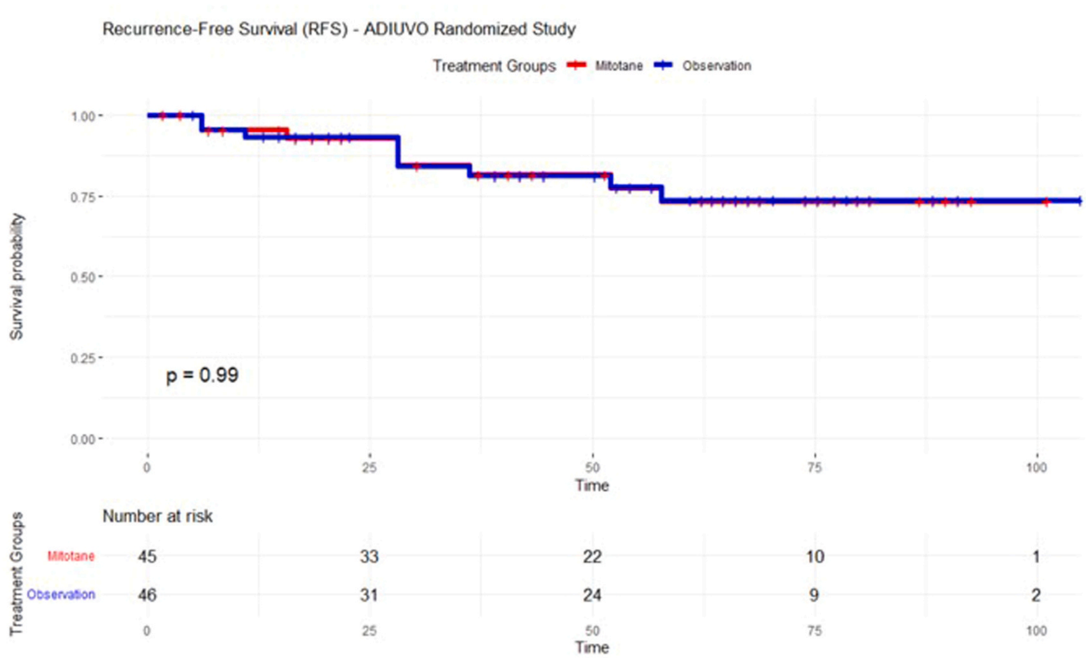

Fig. 1. Recurrence-Free Survival in the randomized ADIUVO trial. KM estimates of RFS for patients with completely resected low-risk ACC randomized to adjuvant mitotane or observation. In the reconstructed IPD (n = 91; 18 events), there was no significant difference between groups (Cox HR 0.99, 95%CI 0.39–2.50; log-rank p = 0.99). RMST differences at 60 and 90 months were minimal and not statistically significant. Numbers at risk are shown below the x-axis.

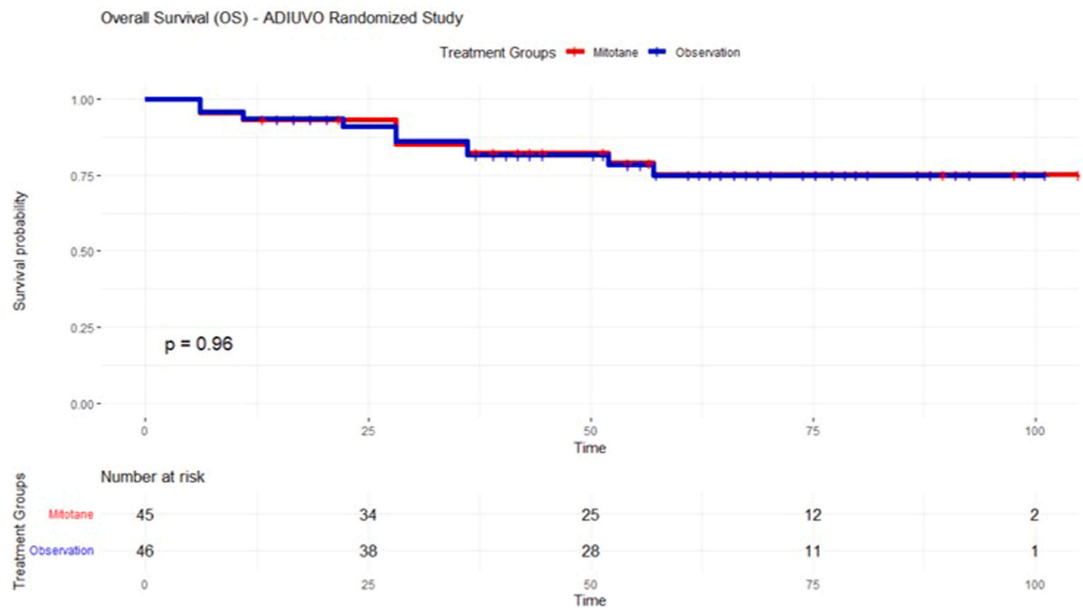

Fig. 2. OS in the randomized ADIUVO trial. KM estimates of OS for patients randomized to adjuvant mitotane or observation. In the reconstructed IPD (n = 91; 19 events), survival curves were nearly superimposable (Cox HR= 0.98, 95%CI; 0.40–2.42; log-rank p = 0.96). RMST differences at 60 and 90 months were negligible and not statistically significant. Numbers at risk are displayed below the x-axis.

mitotane and observation was −0.19 months (95%CI; −6.70–6.32; p = 0.95). At 90 months, the difference was −0.25 months (95%CI −12.08–11.58; p = 0.97). These results indicate that, on average, survival time was essentially indistinguishable between treatment arms across clinically relevant time horizons.

Overall, the reconstructed ITT analysis indicates that no clear evidence of an overall survival advantage with adjuvant mitotane was observed in patients with completely resected, low-risk ACC.

It should be noted that OS and RFS data were reconstructed from separate KM curves and therefore represent endpoint-specific aggregates rather than matched individual patient trajectories. Deaths without prior documented recurrence contribute to OS events but not to RFS events, further explaining the observed discrepancy. Consequently, apparent differences in the number of OS and RFS events reflect differences in endpoint definitions and reconstruction processes rather than inconsistencies in the data.

3.2. *Observational study*

In the reconstructed observational cohort, 15 recurrences were

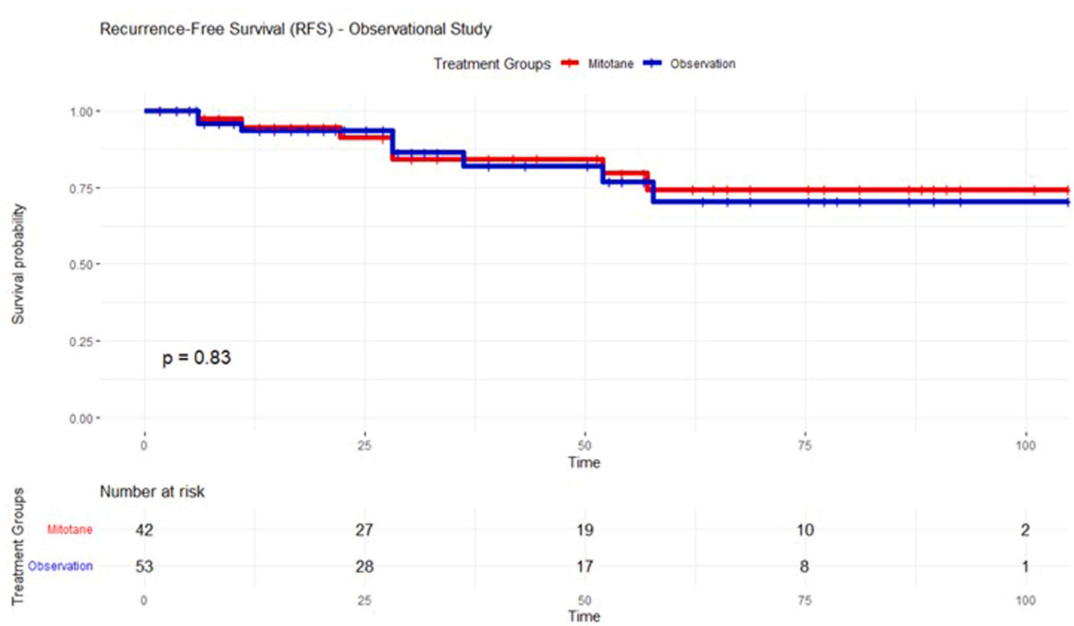

Fig. 3. RFS in the observational cohort (mitotane vs surveillance). KM curves for recurrence-free survival in the prospective observational cohort (n = 95; 15 RFS events). Cox regression showed no significant difference between mitotane and surveillance (HR=0.90, 95%CI; 0.32–2.47; Wald p = 0.83). The log-rank test confirmed the absence of separation between curves (p = 0.83). Absolute differences in RMST were minimal: at 60 months, the between-group difference was 0.43 months (95%CI; −6.62–7.48), and at 90 months, 1.55 months (95%CI; −12.09–15.18), both non-significant.

observed (8 in the surveillance group and 7 in the mitotane group). Cox regression showed no significant difference in recurrence-free survival between groups (HR=0.90, 95%CI 0.32–2.47; Wald p = 0.83). The log-rank test was also non-significant (p = 0.83) (Fig. 3). Restricted mean survival time analysis confirmed minimal absolute differences. At 60 months, the RMST difference (mitotane minus observation) was 0.43 months (95%CI −6.62–7.48; p = 0.91), and at 90 months it was 1.55 months (95%CI −12.09–15.18; p = 0.82).

In the reconstructed observational cohort (n = 95; 20 OS events), no statistically significant difference in OS was observed between mitotane and surveillance. Cox regression yielded a HR of 0.75 (95%CI 0.31–1.84; Wald p = 0.53), indicating a non-significant trend towards lower mortality in the mitotane group. The log-rank test confirmed the absence of a statistically significant difference (p = 0.53) (Fig. 4).

RMST analyses were consistent with these findings. At 60 months, the absolute RMST difference between mitotane and observation was + 1.52 months (95%CI −6.19–9.22; p = 0.70). At 90 months, the difference increased numerically to + 4.50 months (95%CI −8.76–17.77; p = 0.51) but remained non-significant and associated with wide confidence intervals.

Overall, although point estimates numerically favoured mitotane, the effect size was small and statistically non-significant, and confidence intervals were compatible with both modest benefit and no effect.

3.3. *Exposure-response analyses*

In the reconstructed IPD analysis, exposure-response analyses were performed, stratifying the data by mitotane plasma concentrations (target≥14 mg/L vs non-target<14 mg/L) and by early versus late attainment of therapeutic levels. Due to small sample sizes, these analyses were limited in their ability to provide definitive conclusions.

In the time-to-target analysis restricted to patients treated with mitotane, 45 individuals were included (32 in the non-target within 3 months group and 13 in the early-target group), with a total of 8 recurrence events.

Cox regression showed no evidence of improved recurrence-free survival among patients who achieved therapeutic concentrations within three months (HR=1.33, 95%CI 0.32–5.62; Wald p = 0.695). The log-rank test confirmed the absence of difference (p = 0.694) (Fig. 5).

Absolute effect estimates were similarly negligible. At 60 months, the RMST difference between early-target and non-target groups was −0.80 months (95%CI −12.65–11.05; p = 0.90). At 90 months, the difference was −3.36 months (95%CI −22.94–16.23; p = 0.74).

Overall, no clear evidence of improved recurrence-free survival was observed among patients achieving therapeutic mitotane concentrations within three months; however, this analysis was limited by the small sample size and low number of events. The very small sample size (n = 13 in the early-target group) precludes any definitive inference.

In the exposure-based analysis restricted to patients receiving mitotane (n = 42), 25 achieved therapeutic plasma concentrations ≥ 14 mg/L and 17 remained below target. A total of 8 recurrences were observed, 4 in each group. Cox regression showed no significant association between achieving therapeutic concentrations and RFS (HR=0.59; 95%CI 0.15–2.36; Wald p = 0.455). The log-rank test confirmed the absence of a statistically significant difference (p = 0.450).

Absolute treatment effects were similarly negligible. At 60 months, the RMST difference between target and non-target groups was 5.12 months (95%CI −6.88–17.12; p = 0.40). At 90 months, the RMST difference was 7.49 months (95%CI −12.51–27.48; p = 0.46) (Fig. 6). CIs were wide and encompassed both potential benefit and harm.

Overall, although point estimates numerically favoured patients achieving therapeutic mitotane concentrations, the small number of events and broad confidence intervals indicate substantial statistical imprecision. No robust evidence of a clinically meaningful exposure-response relationship for RFS was observed in this low-risk population.

In the reconstructed exposure subgroup (n = 42; 2 OS events), OS did not differ between patients achieving target mitotane levels and those remaining below target. Cox regression yielded an HR of 0.61 (95%CI 0.04–9.79; Wald p = 0.729) and the log-rank test was similarly non-significant (p = 0.726) (Fig. 7). RMST at 60 months showed a minimal absolute difference of 1.05 months (95%CI −5.86–7.96;

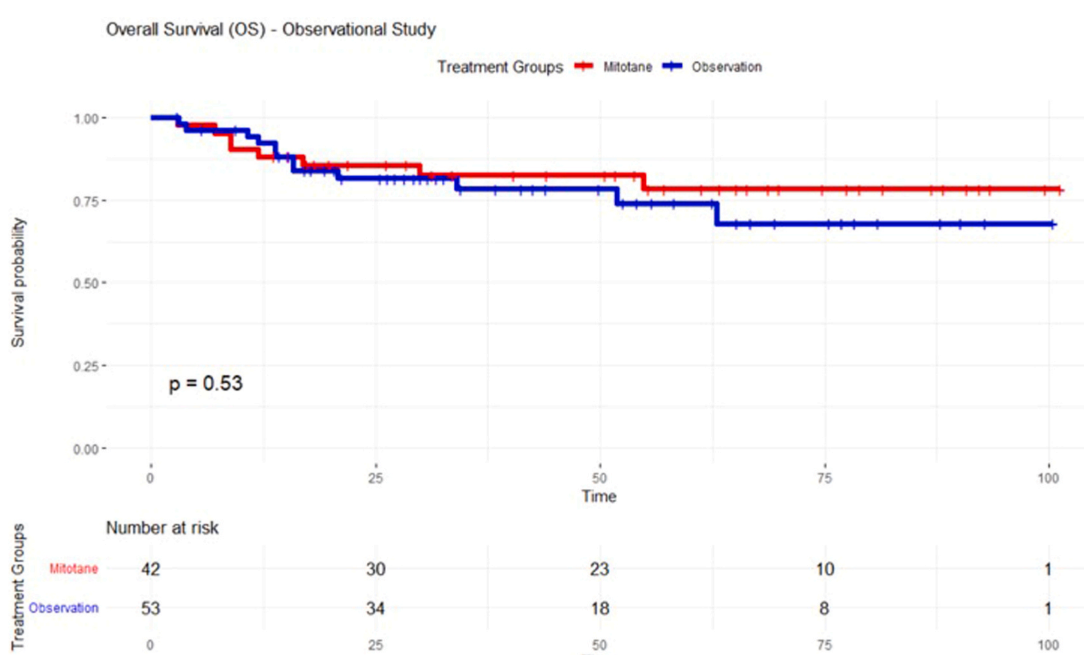

Fig. 4. OS in the observational cohort (mitotane vs surveillance). KM curves for OS in the prospective observational cohort (n = 95; 20 OS events). Cox regression yielded an HR of 0.75 (95%CI; 0.31–1.84; Wald p = 0.53), indicating no statistically significant survival advantage with mitotane. The log-rank test was non-significant (p = 0.53). RMST differences were small and not statistically significant: 1.52 months at 60 months (95%CI; −6.19–9.22) and 4.50 months at 90 months (95%CI; −8.76–17.77).

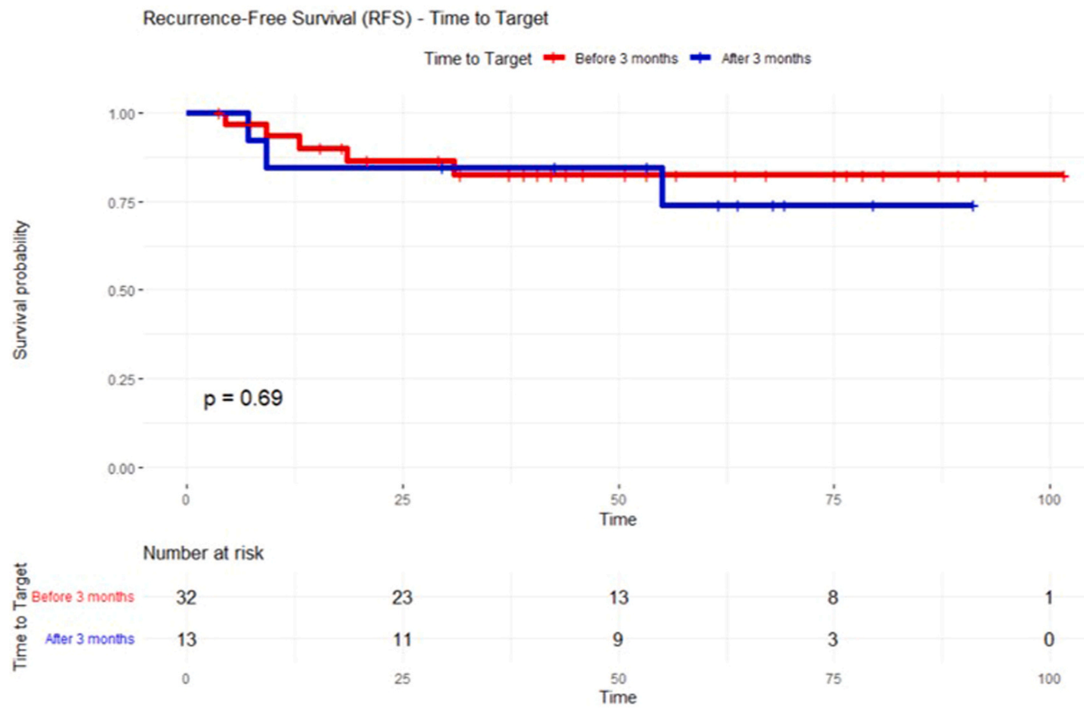

Fig. 5. RFS According to Time to Target Mitotane Concentration. KM curves for recurrence-free survival stratified by time to attainment of therapeutic mitotane plasma levels (≤3 months vs >3 months), after application of a 3-month landmark to mitigate immortal-time bias. Cox regression showed no significant association between early target attainment and recurrence risk (HR=1.33, 95%CI; 0.32–5.62; log-rank p = 0.69). Absolute differences in restricted mean survival time at 60 and 90 months were small and not statistically significant. Numbers at risk are displayed below the plot.

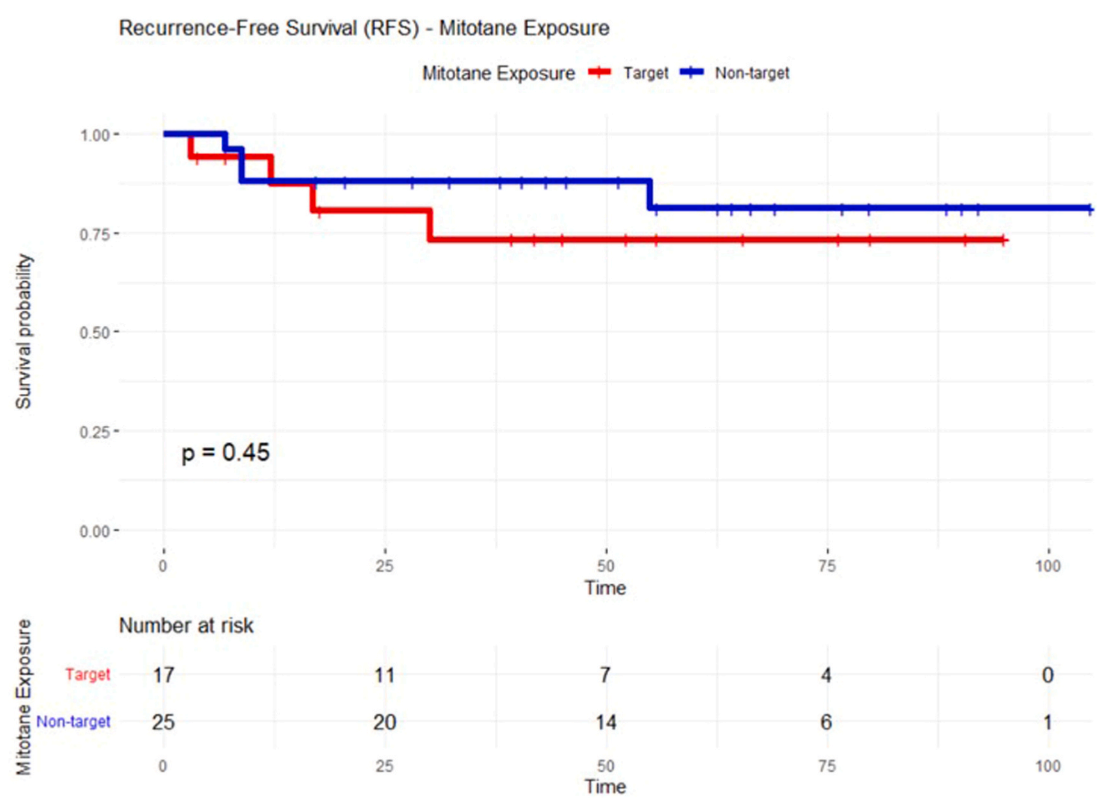

Fig. 6. RFS According to Mitotane Plasma Exposure (target vs non-target). KM curves for RFS comparing patients who achieved therapeutic mitotane plasma concentrations (≥14 mg/L) with those who did not. Cox regression indicated no significant difference in recurrence risk (HR=0.59, 95%CI; 0.15–2.36; log-rank p = 0.45). RMST differences at 60 and 90 months were modest and not statistically significant. Numbers at risk are shown below the plot.

p = 0.766). RMST at 90 months could not be estimated because the maximum follow-up in the non-target group was 78.3 months, making

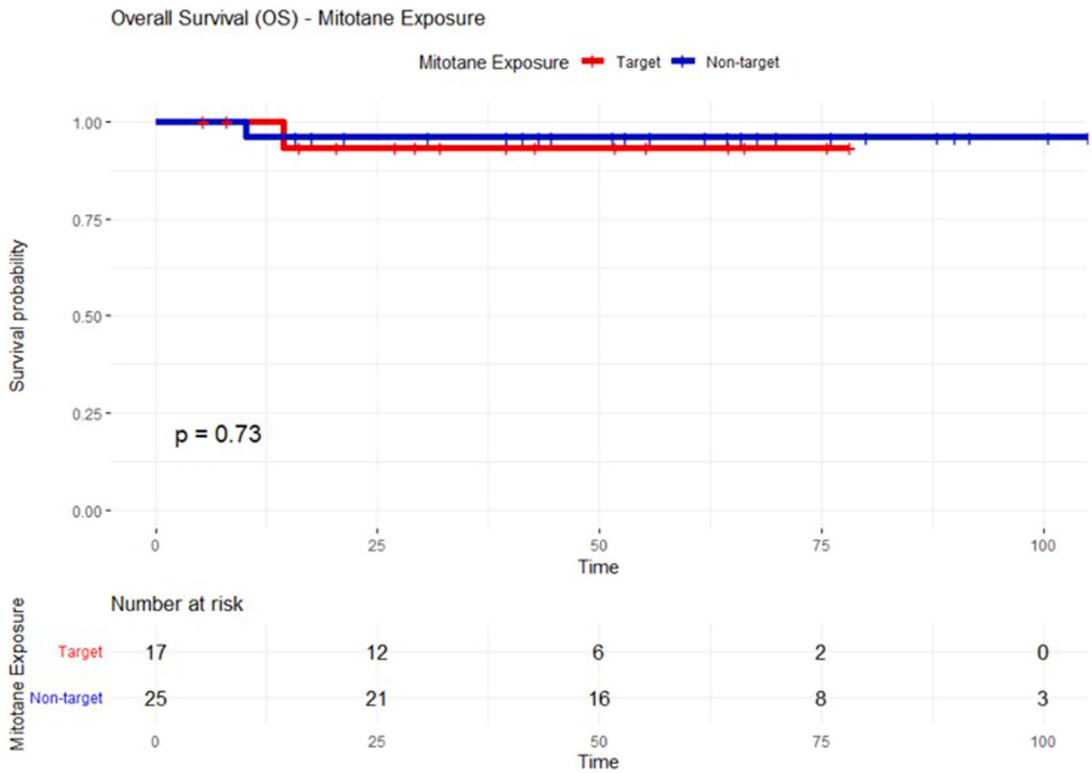

Fig. 7. OS According to Mitotane Plasma Exposure (target vs non-target). KM curves for OS stratified by attainment of therapeutic mitotane plasma concentrations. Cox regression did not demonstrate a significant survival advantage for the target group (HR=0.61, 95%CI; 0.04–9.79; log-rank p = 0.73). RMST at 60 months showed minimal absolute differences, and RMST at 90 months was not estimable due to differential maximum follow-up between groups. Numbers at risk are provided below the plot.

τ= 90 months inadmissible under RMST requirements.

Overall, these subgroup analyses did not identify a consistent signal of benefit associated with higher exposure or earlier target attainment. However, all exposure-based analyses were constrained by small sample sizes and low event counts and should therefore be interpreted with caution. The very small sample size (n = 13 in the early-target group) precludes any definitive inference.

3.4. Pooled randomised and observational analyses

In the pooled IPD analysis combining the randomized and observational cohorts (n = 186), no evidence of a treatment effect emerged for either OS or RFS.

For OS (39 events), the Cox proportional hazards model stratified by study design yielded a HR of 0.86 (95%CI 0.45–1.62; Wald p = 0.63). The corresponding stratified log-rank test was likewise non-significant (p = 0.63), indicating no detectable relative survival advantage with mitotane. The absolute difference in RMST was 1.00 month at 60 months (95%CI −4.07–6.07; p = 0.70) and 2.47 months at 90 months (95%CI −6.45–11.39; p = 0.59), consistent with clinically negligible differences.

For RFS (33 events), the stratified Cox model yielded an HR of 1.05 (95%CI 0.53–2.08; Wald p = 0.895), again showing no statistically significant difference in recurrence risk between treatment groups. The stratified log-rank test was similarly non-significant (p = 0.895). RMST analyses confirmed the absence of a clinically meaningful absolute effect, with differences of −0.18 months at 60 months (95%CI −5.07–4.71; p = 0.94) and −0.53 months at 90 months (95%CI −9.73–8.67; p = 0.91), corresponding to differences of only a few days over clinically relevant follow-up horizons.

Overall, the pooled IPD analysis provides no statistically or clinically meaningful evidence that adjuvant mitotane improves OS or RFS in low-risk ACC when integrating randomized and prospective real-world data. For visual representation, study-specific hazard ratios were also combined using a random-effects meta-analysis, and forest plots were generated to display between-study consistency. Supplementary Figures 1 and 2 present study-specific estimates and pooled hazard ratios for OS and RFS.

These pooled estimates should not be interpreted as causal effects given the non-randomized nature of the observational component. Rather, they provide a complementary assessment of the consistency of treatment effect estimates across different study designs. The close agreement between randomized and observational findings supports the robustness of the overall results despite the potential for residual confounding.

3.5. Censoring patterns and follow-up

In the randomized trial (n = 91), censoring was substantial for both endpoints, consistent with the low event rate in low-risk ACC. For RFS, 18 recurrences occurred (19.8%), while 73 patients were censored (80.2%). Censoring was balanced between treatment arms (80.0% with mitotane vs 80.4% with observation). The reverse KM estimate of median follow-up was 62.3 months (95 CI 51.5–67.6). For OS, 19 deaths were observed (20.9%), with 72 patients censored (79.1%); censoring remained similar across arms (80.0% with mitotane vs 78.3% with observation). Reverse KM median follow-up for OS was 64.7 months (95%CI 57.4–70.3).

In the prospective observational cohort (n = 95), censoring was likewise substantial. For RFS, 15 events occurred (15.8%) and 80 patients were censored (84.2%), with comparable censoring between groups (83.3% mitotane vs 84.9% observation). Reverse KM median follow-up was 39.1 months (95%CI; 27.2–56.7). For OS, 20 deaths occurred (21.1%) and 75 patients were censored (78.9%), again balanced between arms (81.0% mitotane vs 77.4% observation); median follow-up was 52.6 months (95%CI; 41.2–63.3). Overall, censoring patterns were comparable between treatment groups in both the randomized and observational cohorts, with no evidence of systematic imbalance likely to bias hazard-based or RMST estimates.

3.6. AFT models

AFT models were fitted to provide a complementary time-based interpretation of treatment effects by estimating the multiplicative impact of mitotane on survival time (time ratio, TR). Weibull, log-normal, and log-logistic distributions were evaluated in each dataset. Across all analyses, the log-normal model consistently showed the lowest AIC values, indicating the best overall fit.

In the randomized cohort, no evidence of a time-accelerating or time-delaying effect of mitotane was observed. For RFS, the log-normal model estimated a TR of 0.99 (95%CI 0.37–2.63; p = 0.98; AIC=237.4), indicating complete overlap in recurrence timing between treatment arms. For OS, the corresponding TR was 1.04 (95%CI; 0.38–2.85; p = 0.93; AIC=252.5), again showing no meaningful difference in survival time.

In the observational cohort, results were directionally similar. For RFS, the log-normal model produced a TR of 1.13 (95%CI 0.42–3.06; p = 0.81; AIC=199.8), while for OS the estimated TR was 1.40 (95%CI 0.43–4.55; p = 0.57; AIC=254.5). Although the OS estimate numerically favoured mitotane, the wide confidence intervals and non-significant p-values indicate substantial uncertainty and do not support a statistically reliable prolongation of survival time.

In the pooled individual patient data analysis integrating randomized and observational cohorts with study included as a covariate, findings remained consistent. For RFS (33 events), the log-normal model estimated a TR of 1.05 (95%CI; 0.52–2.12; p = 0.88; AIC=433.3). For OS (39 events), the corresponding TR was 1.21 (95%CI 0.56–2.63; p = 0.63; AIC=503.6). These pooled results indicate no evidence that mitotane materially accelerates or delays time to recurrence or death.

Overall, across randomized, observational, and pooled datasets, AFT models consistently produced time ratios close to 1.0 with wide confidence intervals encompassing both potential benefit and harm. The absence of a consistent signal across parametric distributions and study designs mirrors the results obtained with Cox proportional hazards models and restricted mean survival time analyses, reinforcing the conclusion that adjuvant mitotane does not substantially modify the clinical course of low-risk adrenocortical carcinoma. A comprehensive summary of hazard ratios, RMST differences and AFT time ratios across all analyses is provided in Table 1.

4. Discussion

In this comprehensive reanalysis of the ADIUVO program, we employed reconstructed IPD and advanced time-to-event modelling to reassess the role of adjuvant mitotane in patients with completely resected, low-risk ACC. By integrating data from the RCT, the parallel observational cohort, and the exposure-response sub-studies, we created a unified evidentiary framework extending substantially beyond the original analyses. The overall results consistently suggest that, in low-risk ACC, adjuvant mitotane provides at most a negligible clinical benefit in terms of RFS and OS.

Across all analytical approaches, treatment effect estimates remained close to unity. Cox proportional hazards models demonstrated hazard ratios near 1.0 in both the randomized and observational datasets, with non-significant p-values and with wide confidence intervals that do not support a consistent or clinically meaningful survival advantage. Pooled analyses combining randomized and observational cohorts yielded virtually identical estimates, with no meaningful heterogeneity between study components. These findings reinforce and quantitatively strengthen the conclusions of the original ADIUVO trial (Terzolo et al., 2023).

The integration of randomized and observational data represents a key methodological aspect of this study, allowing for a broader assessment of treatment effects. However, this approach also introduces important limitations. In the observational cohort, treatment allocation was determined by physician choice rather than randomization, potentially reflecting unmeasured clinical factors influencing treatment decisions.

Therefore, pooled analyses should not be interpreted as providing causal inference but rather as an assessment of the consistency of treatment effect estimates across complementary data sources. Importantly, the close agreement between randomized and observational findings supports the robustness of the overall conclusions despite these inherent limitations.

Baseline characteristics were reported as broadly comparable between groups in the original ADIUVO study, although residual confounding cannot be excluded.

Importantly, RMST analyses revealed extremely small absolute differences between treatment arms at clinically relevant time horizons. In a population characterized by very low event rates, any potential benefit of mitotane appears to be measured in weeks rather than months. Such marginal gains must be interpreted in the context of treatment-related toxicity and long-term treatment burden (Haak et al., 1994).

Although early follow-up was characterised by instability due to sparse events, HR curves stabilised over time and consistently approached unity in both RFS and OS analyses. No delayed protective effect emerged during longer follow-up, and long-term hazards remained essentially superimposable between treatment arms.

AFT models provided an alternative and complementary perspective by directly modelling survival time rather than hazard. Across randomized, observational, and pooled datasets, time ratios remained close

Table 1
Comprehensive summary of treatment effect estimates across randomized, observational, exposure-based, and pooled analyses. This table summarizes hazard ratios (HRs) from Cox proportional hazards models, absolute restricted mean survival time (RMST) differences at 60 months, and time ratios (TRs) from best-fitting log-normal accelerated failure time (AFT) models across all reconstructed datasets. Comparisons include the randomized intention-to-treat population, the prospective observational cohort, exposure-based subgroup analyses (target ≥14 mg/L vs non-target), time-to-target analyses (≤3 months vs >3 months), and pooled analyses integrating randomized and observational cohorts. Across all endpoints and modelling frameworks, effect estimates remained close to unity with wide confidence intervals and non-significant p-values, indicating no statistically or clinically meaningful survival benefit associated with adjuvant mitotane in patients with completely resected, low-risk adrenocortical carcinoma.

Endpoint / Analysis	ADIUVO (Published) HR (95% CI)	IPD Cox HR (95% CI), p	RMST Δ at 60 m (months)	AFT TR (log-normal) (95% CI), p
RCT - RFS	0.74 (0.30–1.85)	0.99 (0.39–2.50), p = 0.99	-0.05	0.99 (0.37–2.63), p = 0.98
RCT - OS	0.46 (0.08–1.92)	0.98 (95%CI; 0.40–2.42) p = 0.97	-0.19	1.04 (0.38–2.85), p = 0.93
OBS - RFS	0.64 (0.24–1.69)	0.90 (0.32–2.47), p = 0.83	+ 0.43	1.13 (0.42–3.06), p = 0.81
OBS - OS	1.08 (0.22–5.23)	0.75 (0.31–1.84), p = 0.53	+ 1.52	1.40 (0.43–4.55), p = 0.57
RCT - RFS (Time-to-target ≤3 m)	1.34 (0.32–5.67)	1.33 (0.32–5.62), p = 0.69	-0.80	–
RCT - RFS (Mitotane ≥14 mg/L)	0.59 (0.15–2.36)	0.59 (0.15–2.36), p = 0.46	+ 5.12	–
RCT - OS (Mitotane ≥14 mg/L)	0.61 (0.04–9.71)	0.61 (0.04–9.79), p = 0.73	+ 1.05	–
Pooled (RCT+OBS) -RFS	Not reported	1.05 (0.53–2.08), p = 0.90	-0.18	1.05 (0.52–2.12), p = 0.88

to 1.0, with wide confidence intervals and no statistically significant differences. This modelling framework, which does not rely on the proportional hazards assumption, further supports the conclusion that mitotane does not materially alter the natural history of low-risk ACC (Royston and Parmar, 2002).

The exposure-response analyses provide additional biologically relevant insight. One possible explanation for the neutral findings of the ADIUVO trial has been that insufficient drug exposure or delayed attainment of therapeutic mitotane concentrations might have attenuated treatment efficacy (Terzolo et al., 2023; Puglisi et al., 2020). However, our reanalysis does not support this hypothesis. No clear association between achieving target plasma concentrations ($\geq$14 mg/L) and improved outcomes was observed. Although point estimates numerically favoured patients achieving target levels, confidence intervals were wide and compatible with both benefit and no effect. Similarly, early attainment of therapeutic concentrations did not confer a consistent survival advantage, although these findings should be interpreted cautiously given the limited sample size and low number of events. Results were stable across multiple analytical approaches, including relative and absolute measures of treatment effect. Taken together, these findings suggest that the absence of benefit cannot be explained solely by inadequate drug exposure or suboptimal pharmacokinetics but rather reflects the limited impact of mitotane in this favourable-risk population (Terzolo et al., 2023; Cazaubon et al., 2019).

In a population characterized by a low baseline risk of recurrence, even modest relative treatment effects are likely to translate into minimal absolute benefit. This is reflected in the consistently small RMST differences observed across analyses, suggesting that any potential survival gain is likely to be clinically marginal.

From a biological standpoint, these findings are consistent with the known pharmacology of mitotane (Hermsen et al., 2011; Flauto et al., 2024). Mitotane exerts antitumour activity through multiple adrenal-selective mechanisms, including disruption of mitochondrial function and steroidogenesis, and, more recently, putative inhibition of SOAT1 with downstream endoplasmic-reticulum stress in ACC models (Flauto et al., 2024). However, even in advanced disease the objective response rate to mitotane is limited and no biomarker has been consistently validated to identify a clearly mitotane-sensitive biology (Flauto et al., 2024; Sbiera et al., 2015; Terzolo et al., 2013). In this context, the absence of a clear exposure-response signal in the low-risk ADIUVO population suggests that prognosis is driven predominantly by intrinsically favourable tumour biology and low baseline hazard of recurrence, rather than by suboptimal drug exposure. Therefore, strategies aimed primarily at intensifying therapeutic drug monitoring or accelerating dose escalation are unlikely to yield clinically meaningful improvements in long-term outcomes in this specific subgroup, while they may still increase treatment burden and toxicity (Haak et al., 1994; Terzolo et al., 2023).

This study also illustrates the methodological value of reconstructed IPD in rare cancers (Tierney et al., 2007; Wei et al., 2015; Liu et al., 2021). By combining multiple modelling approaches, Cox regression, RMST and AFT analysis, we were able to derive consistent conclusions from complementary statistical perspectives. The convergence of results across these independent frameworks strengthens confidence that the absence of effect reflects biological reality rather than model-specific artefact.

Several limitations warrant consideration. Reconstruction of IPD from published KM curves remain an approximation, although validation against original curves demonstrated excellent concordance. The low number of events limits precision and precludes definitive exclusion of very small treatment effects. Exposure-response analyses were constrained by small subgroup sizes, and unmeasured confounding cannot be excluded in the observational component.

Taken together, these findings suggest that in completely resected, low-risk adrenocortical carcinoma, the biological behaviour of the disease may not provide a sufficient therapeutic window for adjuvant mitotane to exert a measurable impact on recurrence dynamics or survival. In this setting, the balance between marginal, statistically uncertain benefit and well-documented toxicity does not support routine adjuvant use.

5. Conclusion

Across randomized, observational, pooled, and exposure-response analyses, treatment effect estimates consistently remained close to unity, with no clear evidence of a clinically meaningful improvement in recurrence-free or overall survival. While small treatment effects cannot be definitively excluded due to the limited number of events, the overall consistency of findings across multiple analytical frameworks suggests that any potential benefit is likely modest at best.

In this favourable-risk population, where prognosis is intrinsically good, such uncertain and potentially marginal gains must be carefully weighed against the well-documented toxicity and quality-of-life burden associated with prolonged mitotane therapy. Taken together, the available evidence does not support routine adjuvant use in this setting, and active surveillance represents a reasonable and evidence-aligned management strategy following complete resection.

6. Critical view

This study provides a comprehensive reappraisal of adjuvant mitotane in low-risk ACC through a reconstructed IPD framework integrating randomized and prospective observational evidence. In contrast to prior analyses, which have relied on aggregate data or have considered the randomized and observational components separately, our work applies a unified analytical strategy combining Cox proportional hazards models, RMST, and AFT modelling. This multidimensional approach allows for a more nuanced interpretation of both relative and absolute treatment effects, as well as time-based dynamics of survival outcomes.

Our findings consistently indicate the absence of a clinically meaningful survival benefit associated with adjuvant mitotane in patients with completely resected, low-risk ACC. Importantly, the convergence of results across multiple statistical frameworks strengthens the robustness of this conclusion and reduces the likelihood that the observed neutrality reflects model-specific assumptions. Furthermore, by incorporating exposure-response analyses, including therapeutic drug levels and time-to-target attainment, this study addresses a key unresolved question from the original ADIUVO trial and provides evidence that inadequate drug exposure is unlikely to account for the lack of efficacy in this setting.

The present work distinguishes itself from the existing literature in three main aspects. First, it represents, to our knowledge, the first effort to reconstruct and jointly analyse IPD from both the randomized ADIUVO trial and its parallel observational cohort within a single inferential framework. Second, it extends beyond conventional hazard-based analyses by incorporating RMST and AFT models, thereby providing complementary perspectives on treatment effect that are less dependent on proportional hazards assumptions. Third, it integrates exposure-based analyses in a unified survival modelling context, offering a clinically relevant evaluation of pharmacokinetic targets.

The scope of this review is deliberately focused on low-risk ACC, defined by favourable pathological and clinical features following complete surgical resection. It does not aim to address the role of mitotane in high-risk or advanced disease, where the balance of evidence remains different. Similarly, while the methodological framework is extensive, the depth of inference is inherently constrained by the limitations of reconstructed data and the low number of observed events. As such, the results should be interpreted as a high-resolution secondary analysis rather than a replacement for original IPD-based investigations.

From a clinical perspective, this analysis supports a more conservative and individualized approach to postoperative management in low-risk ACC. In a population characterized by a low baseline hazard of recurrence, the marginal and statistically uncertain benefits observed here must be carefully weighed against the well-documented toxicity and impact on quality of life associated with prolonged mitotane therapy. These findings reinforce the role of active surveillance as a reasonable and evidence-aligned strategy in this subgroup.

Overall, this study contributes to the current knowledge of the field by providing an integrated, methodologically robust reassessment of the ADIUVO programme and by clarifying the clinical relevance of adjuvant mitotane in a well-defined low-risk population. It highlights the importance of aligning treatment intensity with underlying disease biology and underscores the need for future research focusing on improved risk stratification and identification of patient subsets more likely to benefit from systemic therapy.

CRediT authorship contribution statement

Fabiano Flauto: Conceptualization, Methodology, Formal analysis, Data curation, Writing - original draft. **Filippo Vitale:** Methodology, Investigation, Writing - review & editing. **Giuseppe Neola**: Data curation, Investigation, Writing - review & editing. **Vincenzo Damiano:** Supervision, Conceptualization, Writing - review & editing

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Declaration of Competing Interest

All authors declare no conflict of interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.critrevonc.2026.105346.

Data Availability

This study is based exclusively on reconstructed individual patient data derived from previously published Kaplan-Meier survival curves of the ADIUVO randomized trial and its parallel observational cohort. All source data are publicly available in the original publication (Terzolo et al., Lancet Diabetes Endocrinol, 2023). No new individual-level clinical data were generated. The reconstructed datasets and R code used for the analyses are available from the corresponding author upon reasonable request.

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Fabiano Flauto, MD, is a Medical Oncology resident at the University of Naples Federico II, Italy. His research focuses on genitourinary and adrenal malignancies, with particular interest in clinical research, advanced statistical methodologies and translational research projects.

Giuseppe Neola, MD, is a Medical Oncology resident at the University of Naples Federico II, Italy, and a Research Fellow at Queen Alexandra Hospital, NHS Trust, Portsmouth, UK. His research interests include genitourinary cancers, clinical data analysis and translational research projects.

Filippo Vitale, MD, is a Medical Oncology resident at the University of Naples Federico II, Italy. His research activity focuses on thoracic oncology, with involvement in clinical research.

Vincenzo Damiano, MD, PhD, is a Medical Oncologist and researcher at the University of Naples Federico II, Italy. His research focuses on adrenal malignancies, head and neck cancers, and thyroid tumors, with extensive experience in both clinical and translational research.